Xiang Cheng 程翔

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Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) undergraduate in **Bioinformatics**, School of Medicine (MED) | Expected Graduation: 2027 (2023 -2027)

The University of California, Berkeley (UCB) 2026.1 -2026.5 exchange & visiting student

CGPA：3.745/4.0 MGPA：3.84/4.0 (Top 10% in School)

• Relevant Coursework: Deep Learning(A), Linear Algebra, Machine Learning, Statistics and Data Analysis, Genomics Analysis, Molecular Simulation(A), Molecular Biology

Honors and Skills

- Silver Medal, Chinese Biology Olympiad for High School Students (2022)
- Dean's List, School of Medicine, CUHK-Shenzhen (2023–2025)
- Full Entrance Scholarship, CUHK-Shenzhen (2023–2026; RMB 140,000/year)
- Undergraduate Research Award (2025-2026; RMB 1,000/month)
- Second Prize (Provincial Level), China Undergraduate Mathematical Contest in Modeling (2024)
- Gold Medal, International Genetically Engineered Machine (iGEM) Competition (2025)

Technical skills:
Programming & Data Science: Python, R; foundational deep learning and machine learning workflows
Computational & Systems Tools: Linux, Git
Web Development: HTML (basic)
Biological Foundation: molecular and cellular biology experimental fundamentals; systems-level biological reasoning
Languages: Mandarin Chinese (Native)，English

Research Experience

• **Deep Evolutionary Regulatory Transformer (DERT) Cross-species** | ***Research Project*** **2026.2-now**

We present a preliminary experiment for training a regulatory representation model using multi-modal genomic signal tracks. The model aims to capture latent regulatory states and support downstream functional interpretation of genomic regions.

• **miRNA Features Drive Cross-Task Improvements in Drug Mechanistic Modeling** | ***Research Project***
2024.10-2025.12

Elucidating drug mechanisms of action typically relies on high-dimensional transcriptomic signatures, yet this approach often overlooks intermediate regulatory layers and is prone to overfitting due to the disparity between noisy features and limited sample sizes. By evaluating these features across causal pathway reconstruction and mechanism of action prediction tasks, we demonstrate that integrating regulatory layers acts as a form of biological regularization. This integration significantly mitigates overfitting and enhances model generalization on unseen data.

• **Engineering an AND-Gate Genetic Circuit in E. coli for Anticancer Applications** | **(iGEM 2025, Gold Medal)**
Core Dry-lab Contributor | **Apr 2025 – Oct 2025**

- Led dry-lab modeling and design of an AND-gate plasmid system using SnapGene.
- Built mathematical/computational models for plasmid expression dynamics, bacterial metabolism, and downstream anticancer efficacy.
- Contributed to project website development and technical content architecture.

• **Improved YOLOv5-based Food Recognition for Nutritional Analysis** | ***Course Project*** | **Dec 2025**

- Constructed a self-curated cafeteria food image dataset and trained an improved YOLOv5 model.
- Implemented lightweight deployment for food recognition and nutrition-oriented analytical output.